Unit: Unit 3: Functions
Topic: Composite Functions

Name: __________
Date: ________

Advanced (Habanero)

Q1. Evaluate $f(g(x))$ where $f(x) = x^2$ and $g(x) = 2x+1$. What is $f(g(2))$?
- (A) 15
- (B) 25
- (C) 30
- (D) 36
- (E) 50

Q2. Find the domain of $f(g(x))$ where $f(x) = \frac{1}{x}$ and $g(x) = x - 2$. What is the domain?
- (A) $(-\infty, 2)$
- (B) $(-\infty, 2]$
- (C) $[2, \infty)$
- (D) $(-\infty, 2) \cup (2, \infty)$
- (E) $(-\infty, \infty)$

Q3. Given $f(x) = x^3$ and $g(x) = \sqrt{x}$, evaluate $f(g(4))$. What is the result?
- (A) 6
- (B) 8
- (C) 16
- (D) 24
- (E) 32

Q4. Determine the domain of $f(g(x))$ where $f(x) = \sqrt{x}$ and $g(x) = x^2 - 4$. What is the domain?
- (A) $[-2, 2]$
- (B) $[-2, 2)$
- (C) $(-2, 2)$
- (D) $[-2, \infty)$
- (E) $[2, \infty)$

Q5. Interpret $f(g(x))$ in the context of a population growth model where $f(x) = 2x$ and $g(x) = 3x + 1000$. What does $f(g(10))$ represent?
- (A) Initial population size
- (B) Population after 10 years
- (C) Growth rate after 10 years
- (D) Total population after 2 growth periods
- (E) Average population size over 10 years

Q6. Evaluate $f(g(x))$ where $f(x) = e^x$ and $g(x) = \ln(x)$. What is $f(g(e))$?
- (A) e
- (B) e^e
- (C) $\ln(e)$
- (D) $\ln(e^e)$
- (E) 1

Q7. Find the range of $f(g(x))$ where $f(x) = \sin(x)$ and $g(x) = 2x + 1$. What is the range?
- (A) $[-1, 1]$
- (B) $[-\frac{1}{2}, \frac{1}{2}]$
- (C) $[0, \pi]$
- (D) $[-\pi, \pi]$
- (E) $[0, 1]$

Q8. Determine the domain of $f(g(x))$ where $f(x) = \frac{1}{x-2}$ and $g(x) = x^2 - 4$. What is the domain?
- (A) $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$
- (B) $(-\infty, -2) \cup (2, \infty)$
- (C) $[-2, 2]$
- (D) $(-2, 2)$
- (E) $[2, \infty)$

Open Ended Questions (FRQ)

Q9. Prove that $f(g(x))$ is an even function where $f(x) = x^2$ and $g(x) = -x$. Show all steps.

Q10. Find and interpret the meaning of $f(g(x))$ in a real-world context, such as population growth, where $f(x) = \frac{1}{x}$ and $g(x) = 2x$. Explain the process and final answer.

Chef's Tips

- Tip 1: Ensure strict adherence to the format.
- Tip 2: Maintain concise questions and answers.
- Tip 3: Double-check all mathematical expressions for accuracy.